

Reminders

Upcoming due dates

Wed Data checkpoint

Fri Discussion lab 6

For people who couldn't use canvas quiz Fri you may do your last weeks quiz in any section this week (including just taking double quizzes in your own section)

Machine Learning II

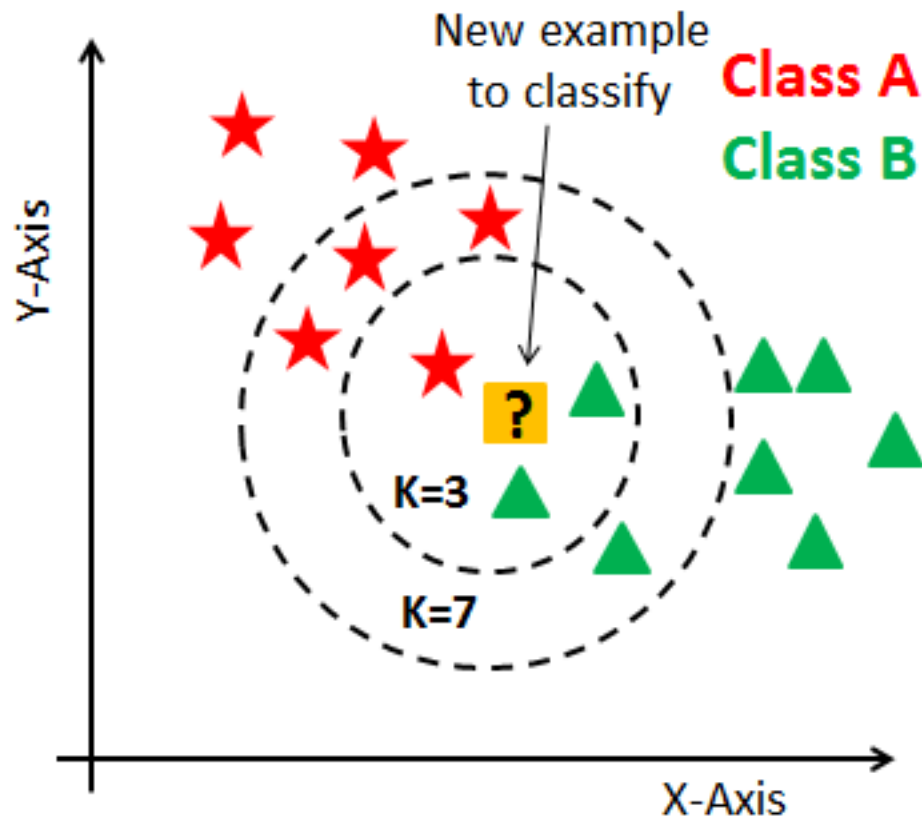
Data Science in Practice

Jason G. Fleischer, PhD

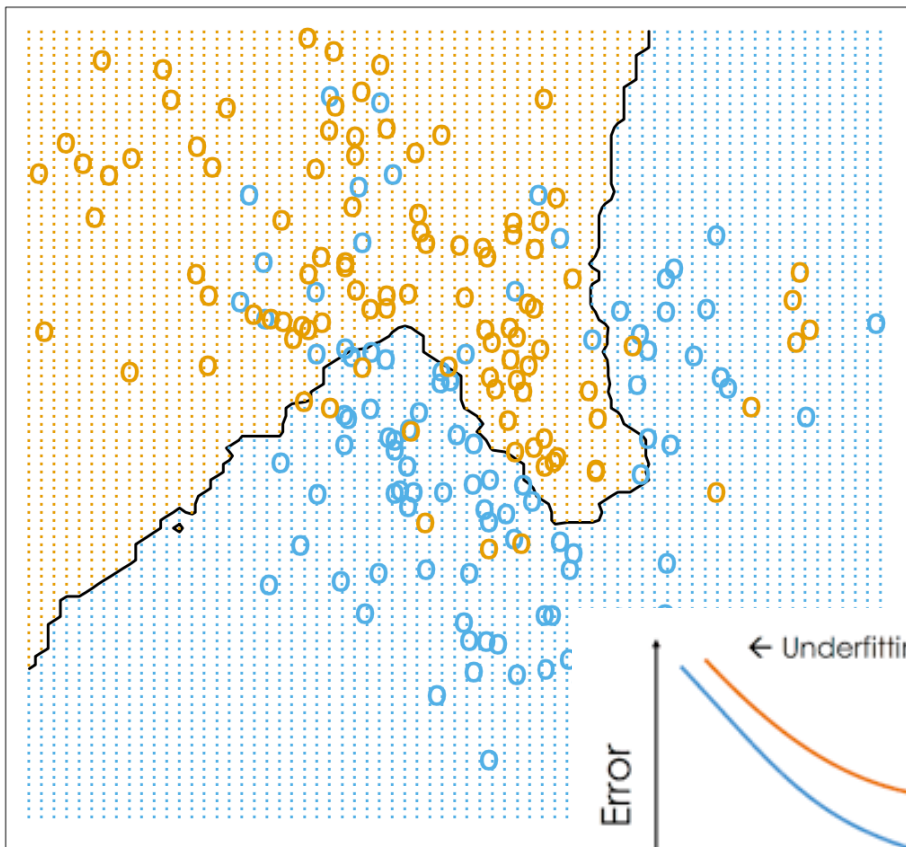
Dept. of Cognitive Science

UC San Diego

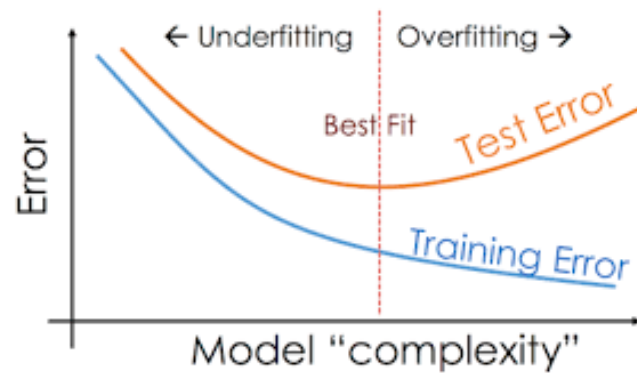
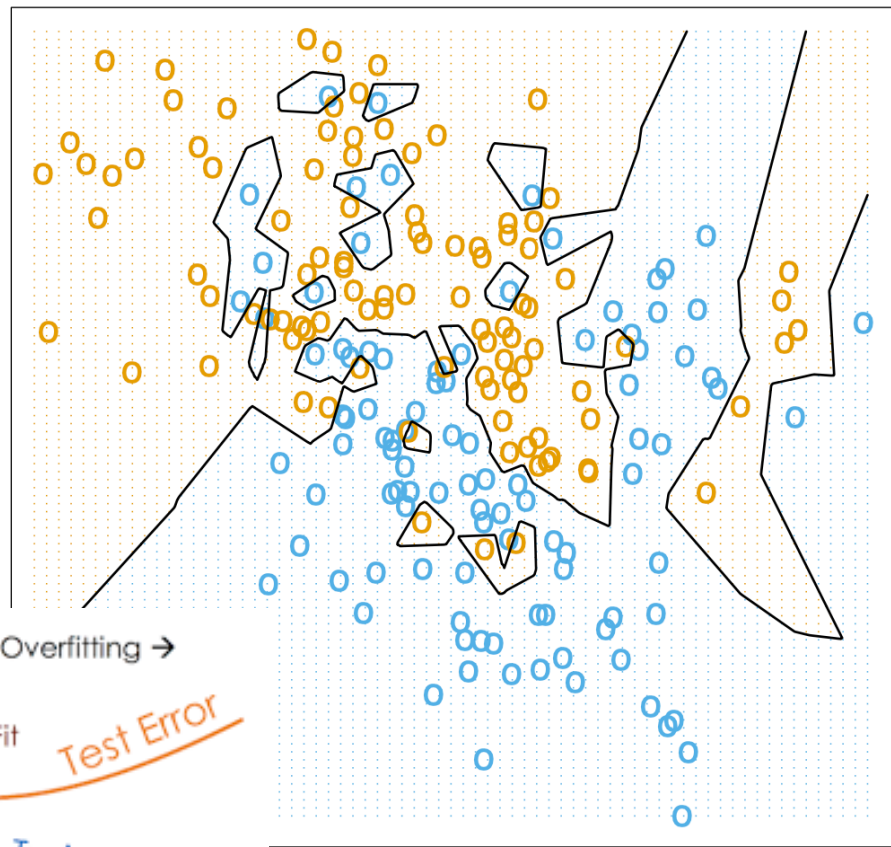
K-nearest neighbors



15-Nearest Neighbor Classifier

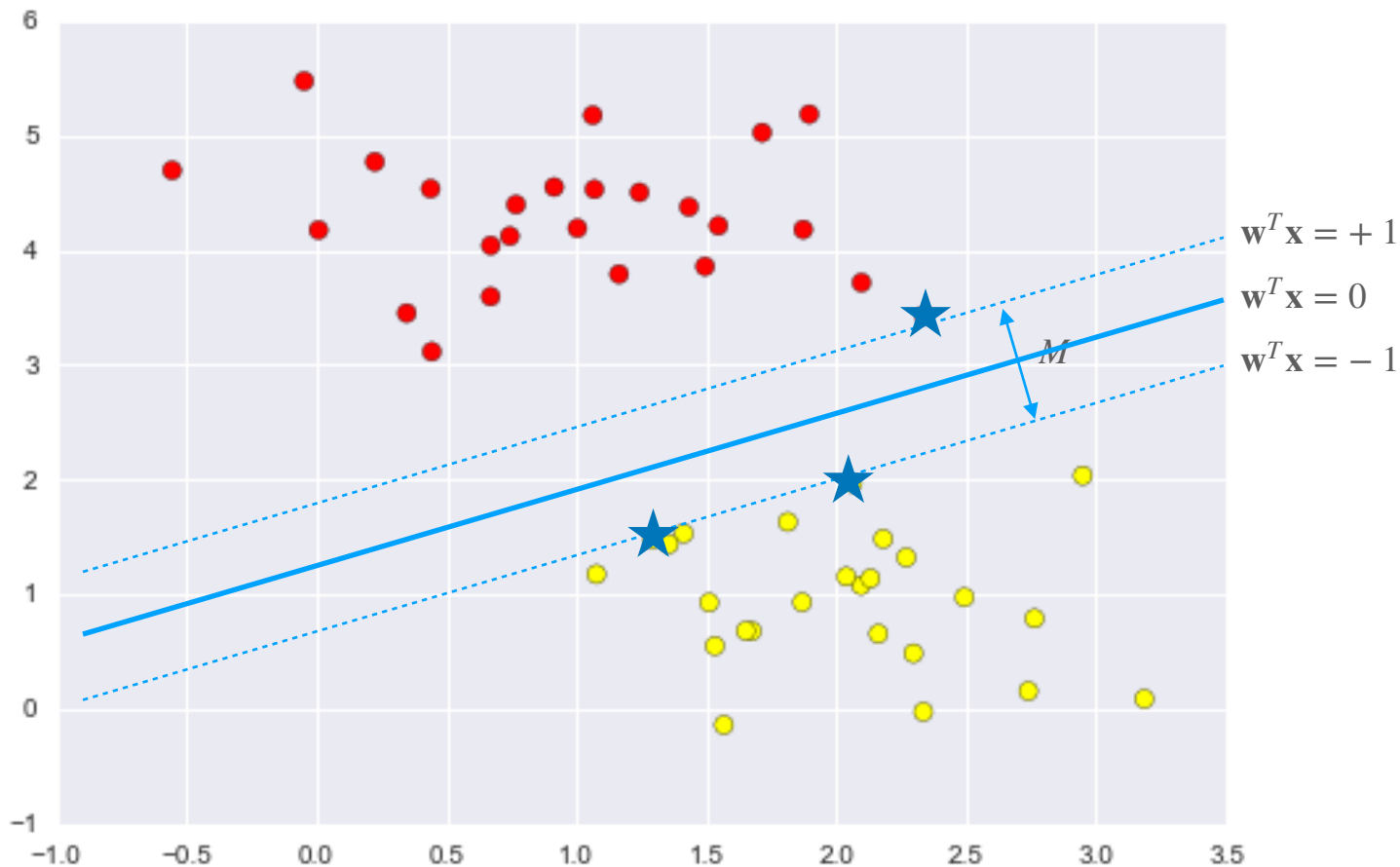


1-Nearest Neighbor Classifier



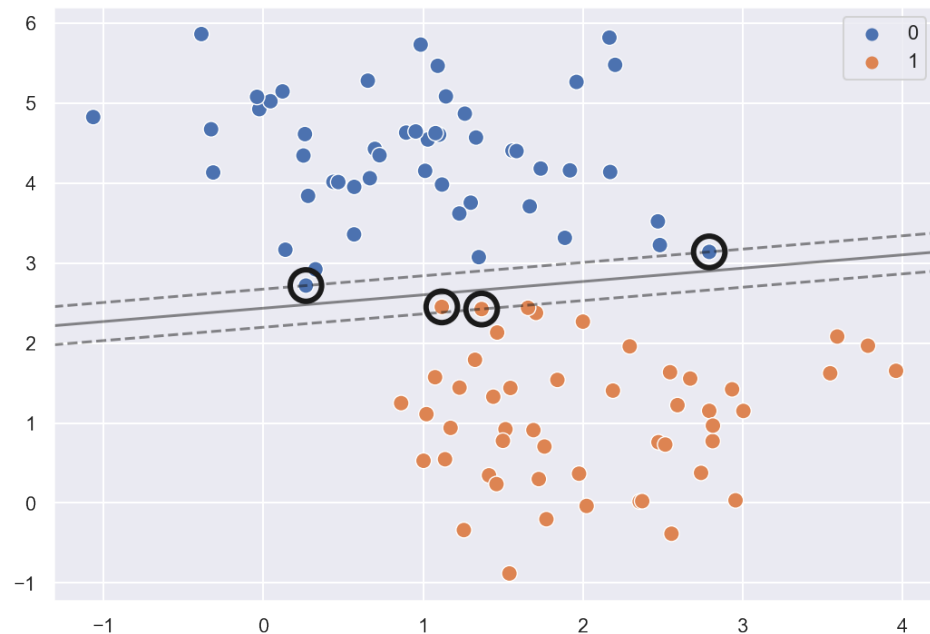
Support Vector Machines:

Find the support
data points that
maximize the
margin M



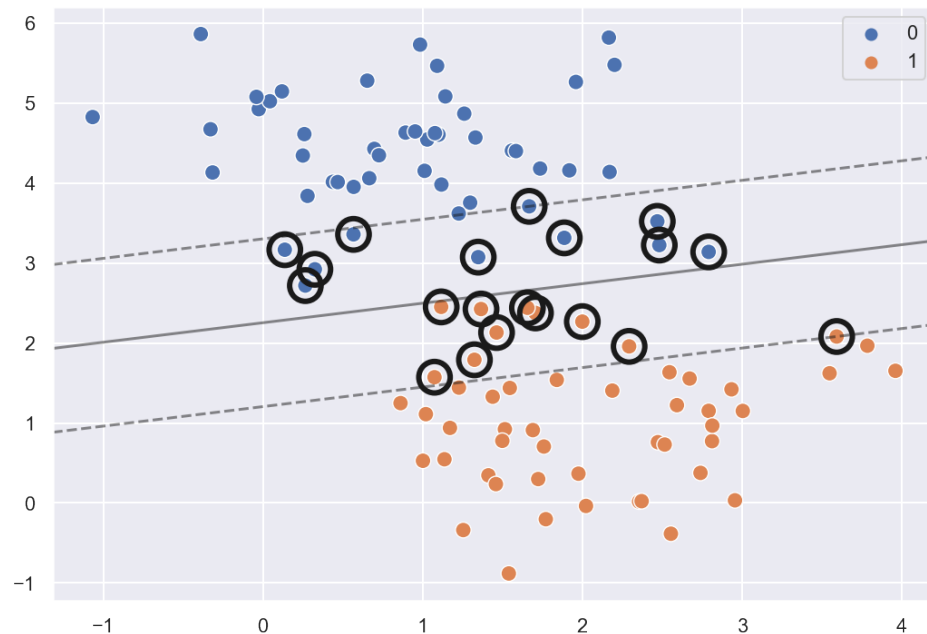
Hard margin

$C = 10.0$



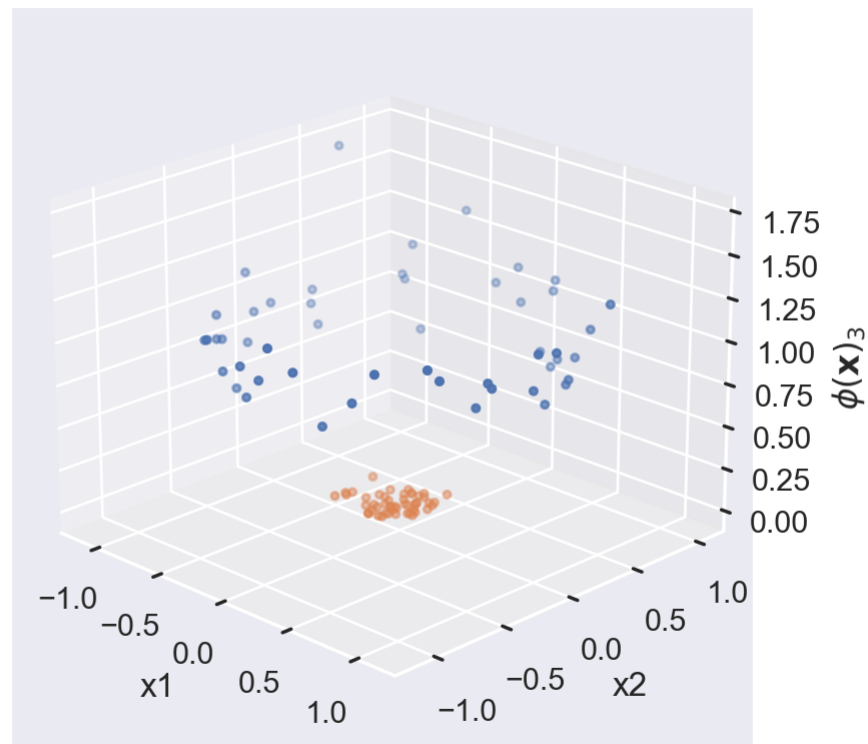
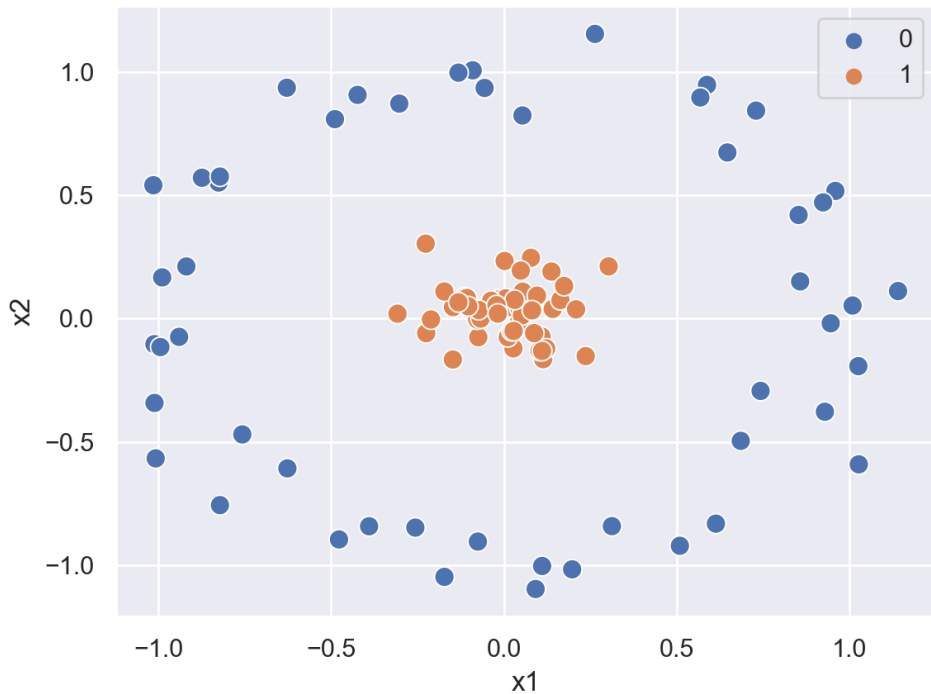
Soft margin

$C = 0.1$

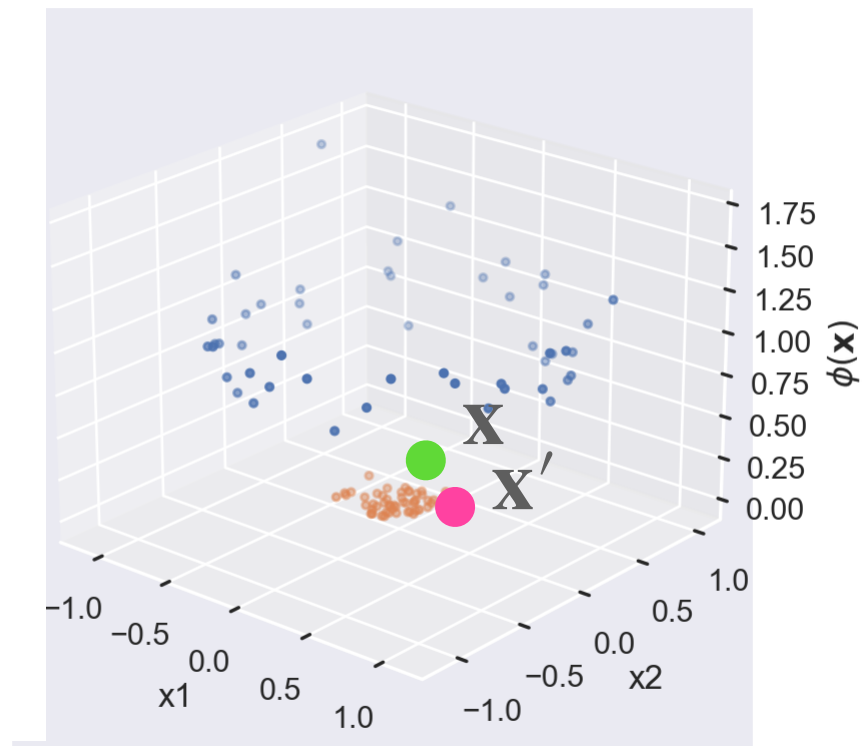
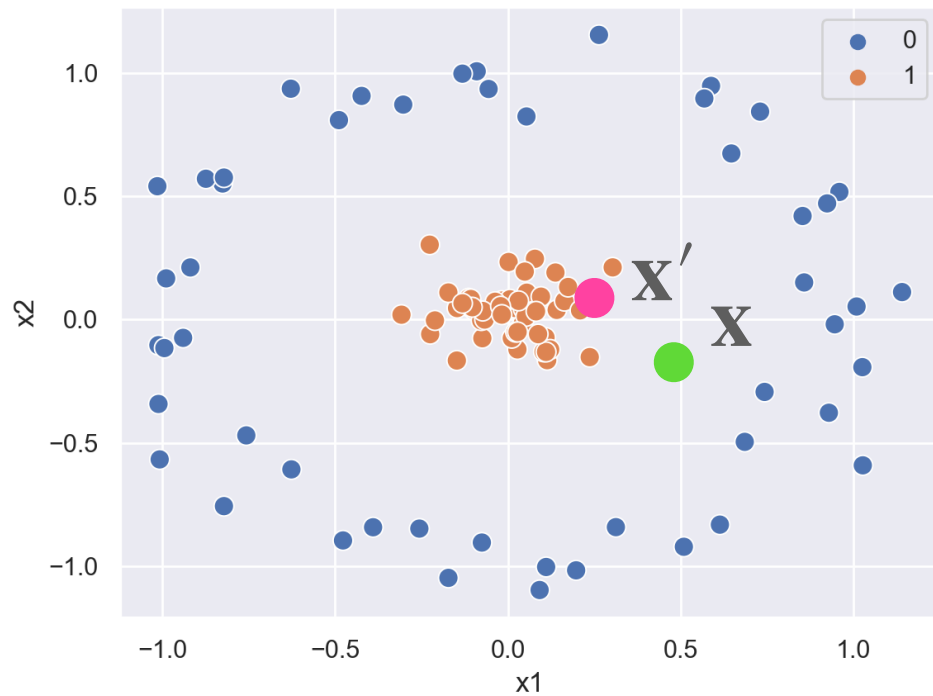


Find: $\arg \min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 + C \times \sum_{i=1}^n (1 - y_i \times (\mathbf{w}^T \mathbf{x}_i + b))_+$

$$\phi(\mathbf{x}) = [x_1, x_2, x_1 \cdot x_2 + x_1^2 \cdot x_2^2]$$



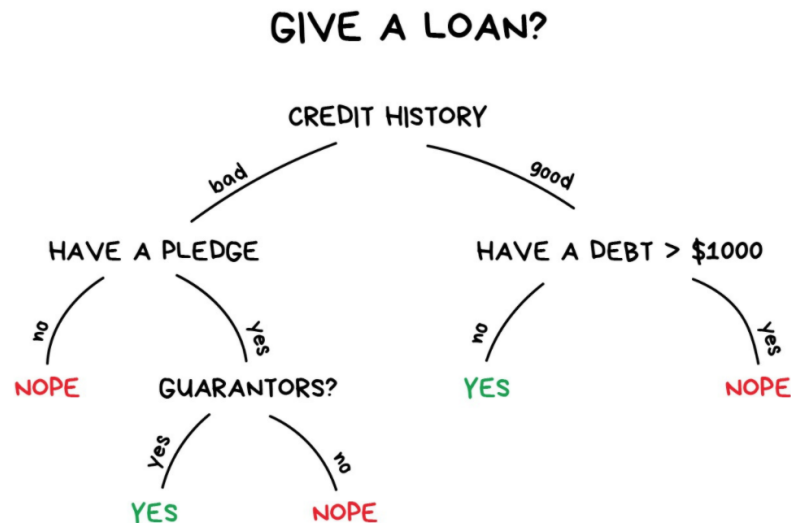
More generally this is the polynomial kernel

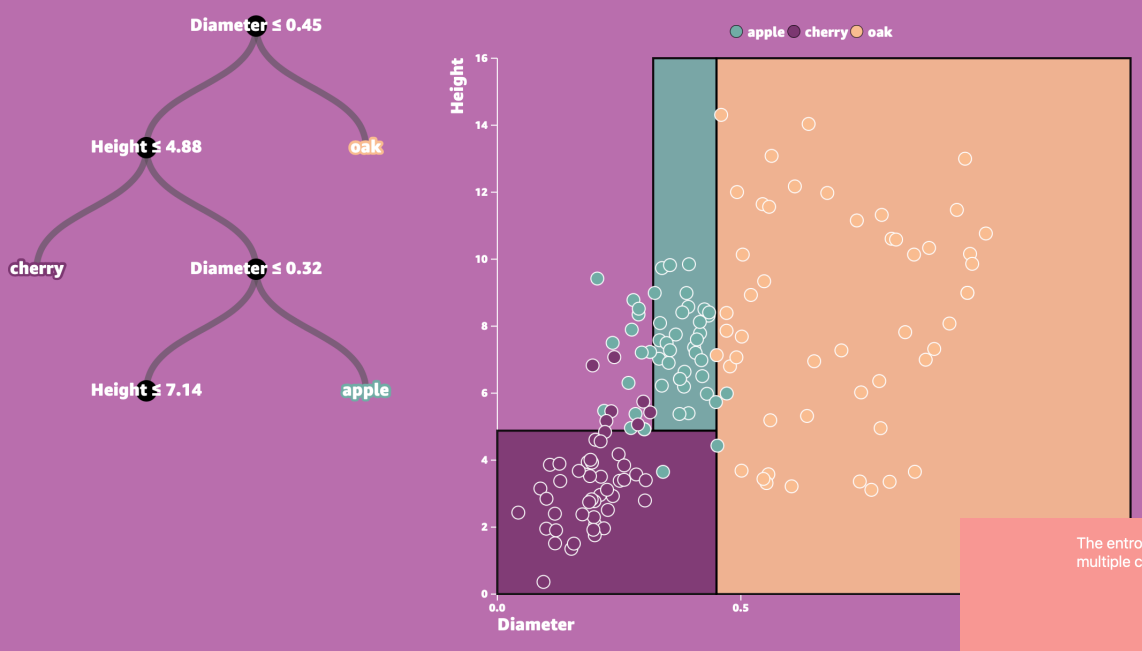


Decision trees

ML that's human interpretable!

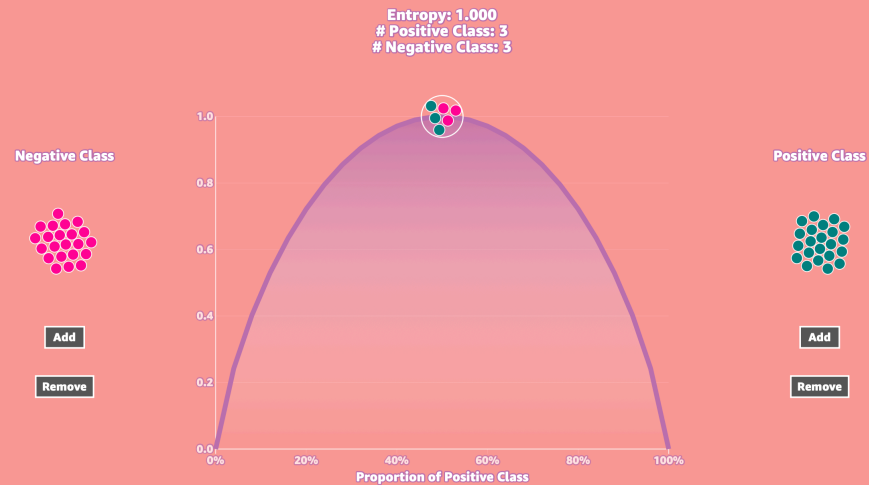
- Each node is
 - an if-then statement
 - visualizable as a decision boundary perpendicular to one axis in the data space
- Nodes arranged in a tree structure
- The axis and its threshold is determined by exhaustive search looking for the choice that minimizes the impurity of nodes after the split
- A tree can be seen as a collection of rules, it can be read by humans

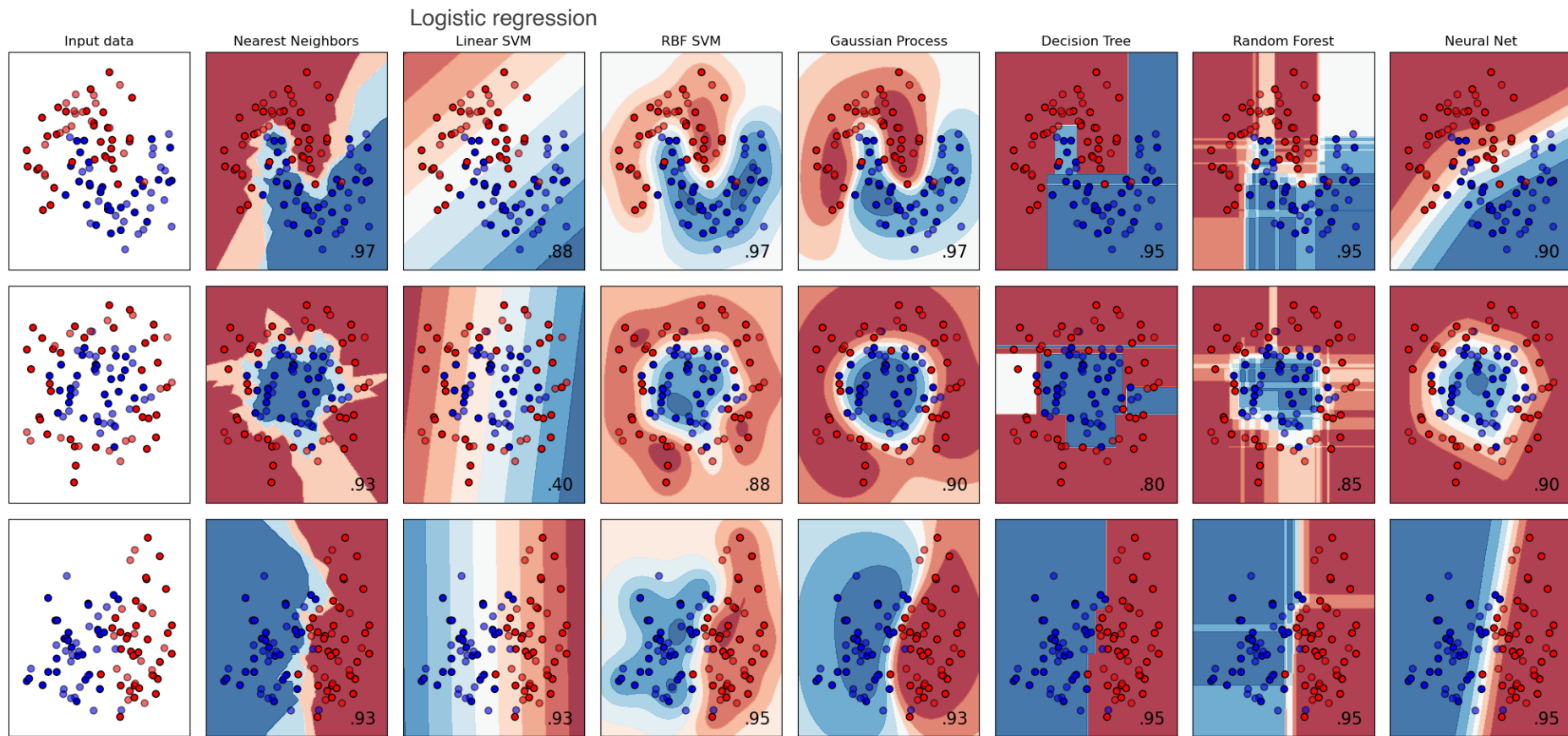




<https://mlu-explain.github.io/decision-tree/>

The entropy can be used to quantify the **impurity** of a collection of labeled data points: a node containing multiple classes is impure whereas a node including only one class is pure.





Linear vs Non-linear classifiers

- LINEAR - Lower complexity, less overfitting, less power
 - Logistic regression
 - “Linear kernel” SVM
- NON-LINEAR - Higher complexity, more overfitting, more power
 - K-nearest Neighbors
 - “RBF kernel” SVM



model assessment

Squared Error

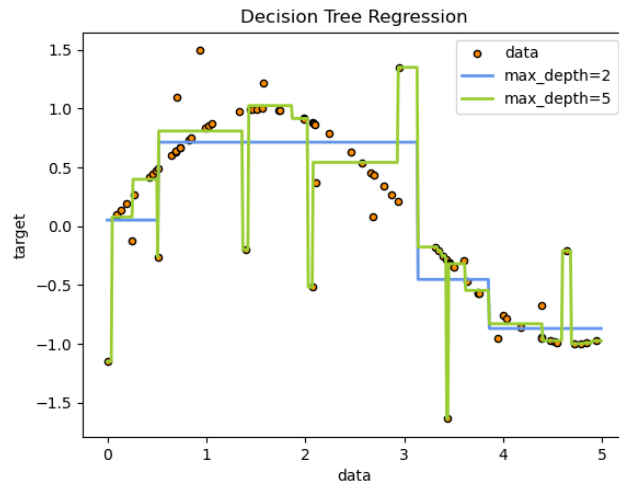
Computationally simpler

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{Y}_i \right)^2$$

Same units as y

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

A few outliers can lead to a big increase in squared errors...
even if all the other predictions are pretty good



Alternatively

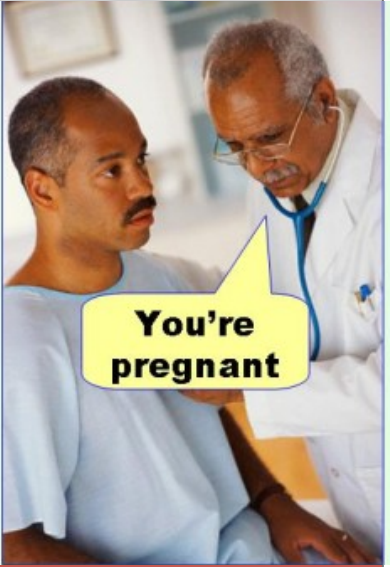
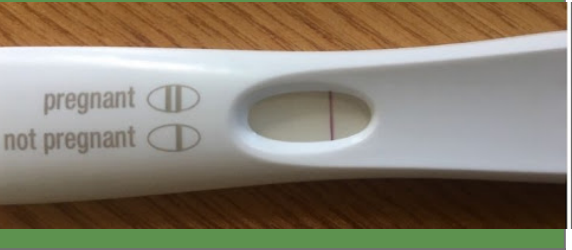
$$\text{MAE} = \frac{\sum_{i=1}^n |y_i - x_i|}{n} = \frac{\sum_{i=1}^n |e_i|}{n}.$$

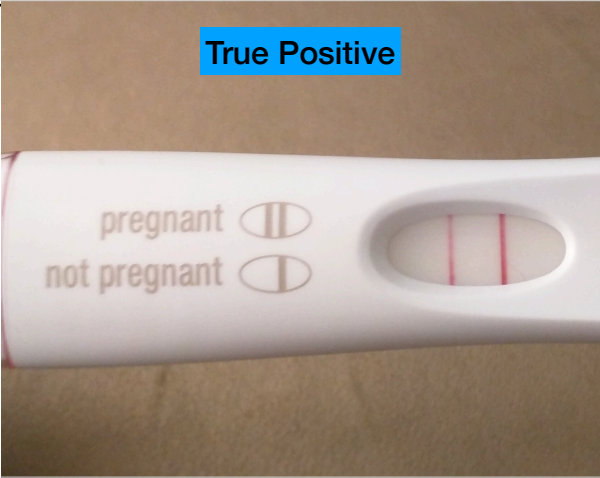
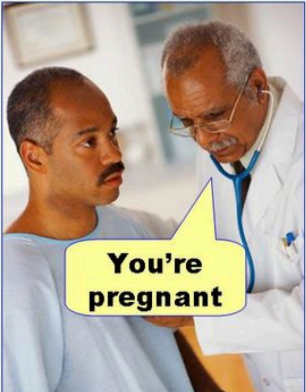
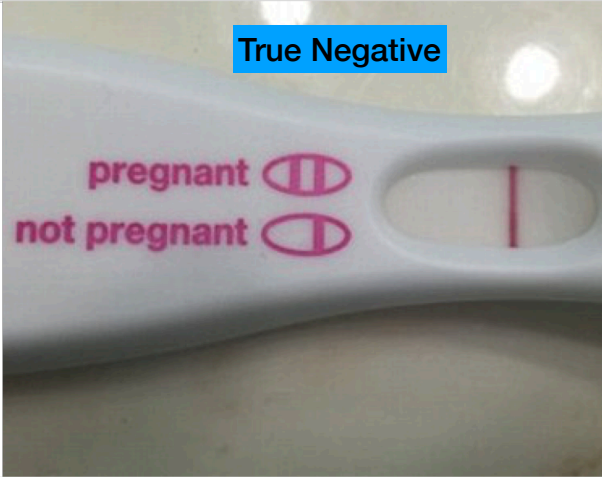
If you care more about most errors, and less about the outliers

$$\text{Accuracy} = \frac{\# \text{ of samples predicted correctly}}{\# \text{ of samples predicted}} * 100$$

Accuracy can mislead

- If classes are imbalanced, what would “chance performance” be?
- The classic example: detect cancer
 - 1/1000 actually have cancer
 - your prediction algorithm misdiagnoses 1% of healthy people
 - Do the math... your algorithm tells 10 people who are healthy they are sick for every 1 person who is actually sick

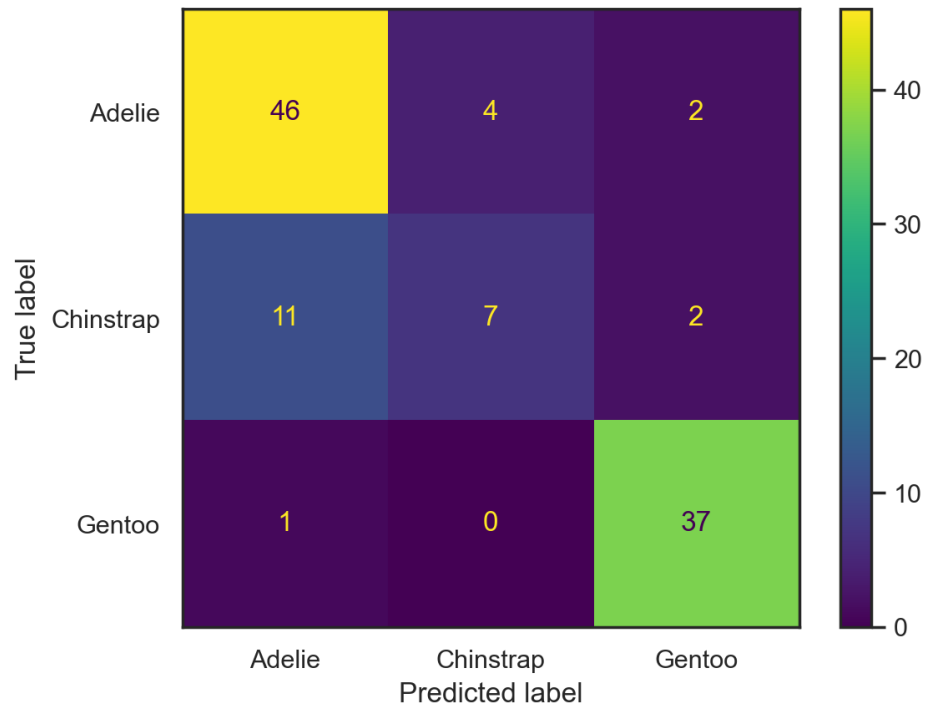
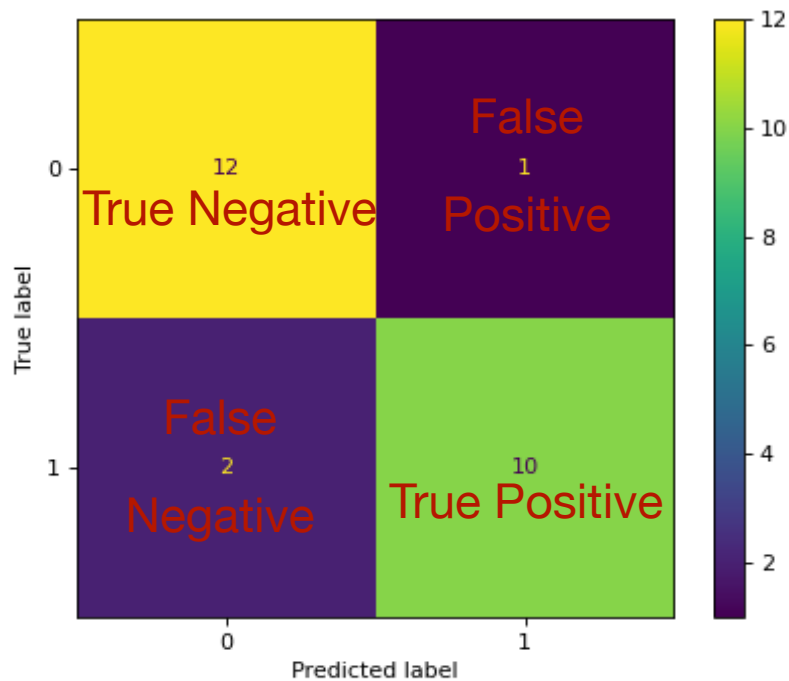
		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP) 	False Positive (FP) 
	Negative	False Negative (FN) 	True Negative (TN) 

	Predicted +	Predicted -
Truly +	True Positive 	Type II error (false negative) 
Truly -	Type I error (false positive) 	True Negative 

confusion matrix

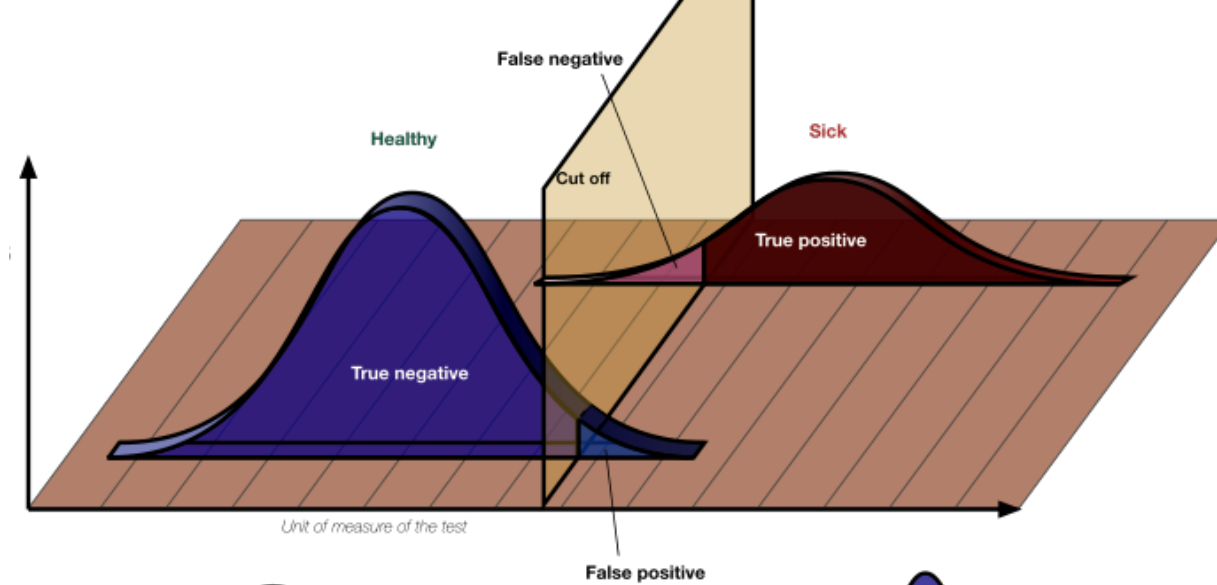
		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Confusion matrix



Accuracy	What % were predicted correctly?
Recall a.k.a. Sensitivity	Of those that <i>were</i> positives , what % were predicted to be positive?
Specificity	Of those that were negatives , what % were predicted to be negative?
Precision a.k.a. PPV	Of those that we predicted positive, what % were positives ?

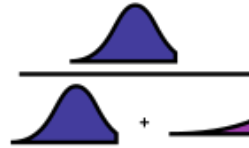
categorical variable prediction



PPV =
aka recall



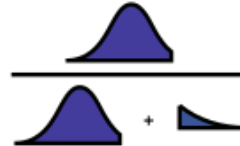
NPV =



Sensitivity =
aka recall



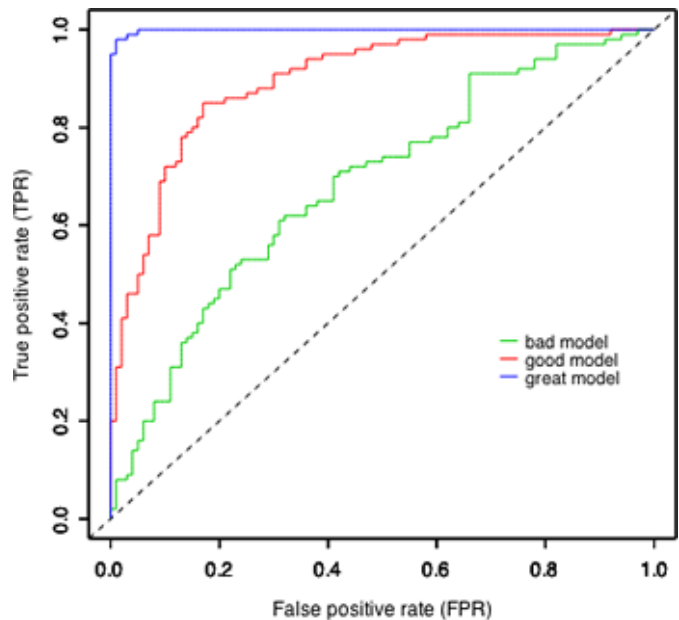
Specificity =



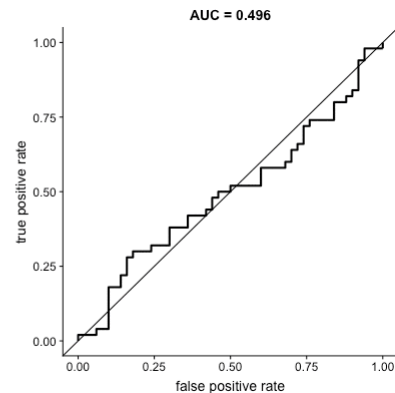
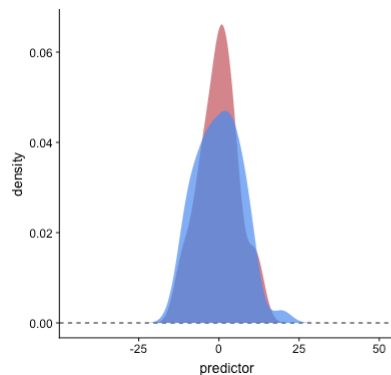
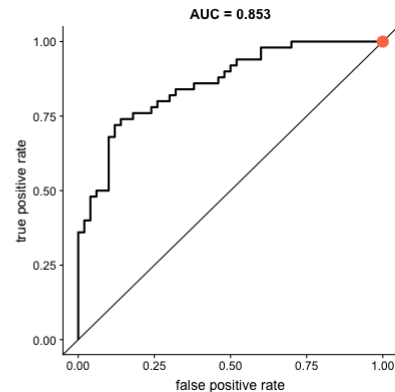
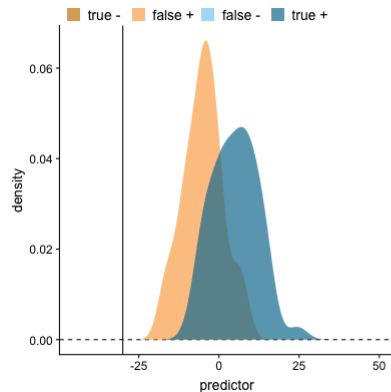
		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Receiver Operating Characteristic

ROC-AUC



The value can range from 0 to 1. However AUC score of a random classifier for balanced data is 0.5

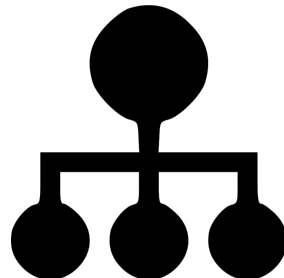
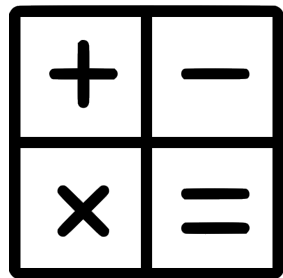
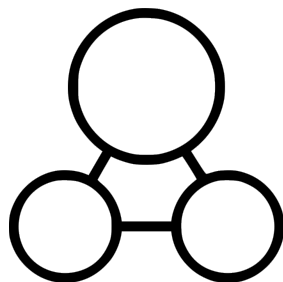




Prediction Approach



[https://forms.gle/
ubN7LFy3FdbSmcEi6](https://forms.gle/ubN7LFy3FdbSmcEi6)



model selection

OUR GOAL IS TO PREVENT OVERFITTING!!!!

Selecting the right model form

Things we can select

- Try different algorithms: k-NN, SVM-linear, Neural network, ...
- For each algorithm try a variety of hyperparameters: $k=1,3,5,7,9\dots$
- For each algorithm try different features: all possible subsets of n variables
- Or any subset of the above!

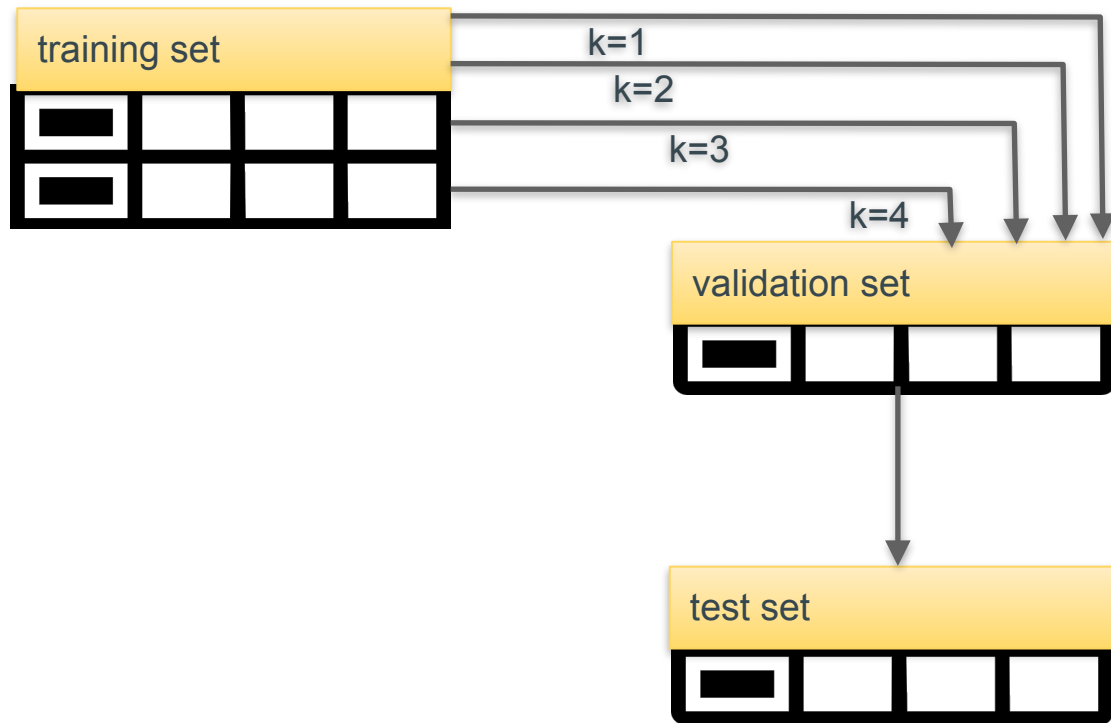
Selecting the right model form

The most common thing is selecting hyperparameters

- Hyperparameter selection
 - k-NN: $k=1,3,5,7,9,11,\dots$
 - You pick k based in part on the dataset. 300 samples and 10 categories? Don't go very high. 300k samples and 2 categories? You can go very big k !
 - SVM: $C=0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000$
 - Your dataset may inform here too... noisy problem, you need soft C values (<1)
 - Decision tree: $\text{max_depth} = 2, 3, 4, 5, 6, 7\dots$
 - Note that number of tree nodes are $2^{\text{max_depth}}$ so this is an exponential progression in complexity just like in SVM above

Selecting a model

Imagine we are using a ML model with a hyper parameter k taking values from 1 to 4



Pick the model that performs best on validation data...

Now retrain that k -value on a combo of training set + validation set

And evaluate its future performance level on the test set